Understanding Household Environment and its linkages with selected Diseases among Elderly Population in India

By

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Declaration

I hereby declare that the work presented in the thesis entitled "Understanding Household Environment and its Linkages with Selected Diseases among 45+ population in India " has been carried out entirely by me under the guidance of Dr. Aparajita Chattopadhyay, Professor, Department of Population and Development, International Institute for Population Sciences, Deonar, Mumbai.

I further declare that this work is original and has not been submitted in part or whole to any university/Institute for the award of any Degree/Diploma.

Date: - April 8, 2024

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I certify that the above declaration is true to the best of my knowledge and belief.

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Dedicated to my family.

Chapter 1. Introduction

"*Health is not just being diseases free. Health is when every cell in your body is bouncing with joy*."

– Sadhguru

1.1 **Background:**

Indoor air pollution is the degradation of indoor air quality by harmful chemicals and other materials; it can be up to 10 times worse than outdoor air pollution. This is because contained areas enable potential pollutants to build up more than open spaces. Statistics suggest that in developing countries, health impacts of indoor air pollution far outweigh those of outdoor air pollution. Indoor air pollution from solid fuels accounted for 3.5 million deaths and 4.5% global daily-adjusted life year (DALY) in 2010; it also accounted for 16% particulate matter pollution. (1) Nowadays, 90% of the rural households in the most developing countries and around 50% of the world's population make use of unprocessed biomass for open fires and poorly functioning cooking stoves indoors. These deficient methods of cooking are responsible for indoor air pollution (IAP). (2) Constant exposure to IAP due to the combustion of solid fuels is the common cause of several harmful diseases in developing countries. The list includes chronic obstructive pulmonary disease (COPD), otitis media, acute respiratory infections, tuberculosis, asthma, lung cancer, cancer of larynx and nasopharynx, low birth rate, perinatal conditions and severe eye diseases that can even cause blindness.

Indoor pollution encompasses a myriad of sources, each posing distinct threats to human health and well-being. Inadequate sanitation facilities contribute to microbial contamination, while dampness fosters mold growth, releasing allergens and toxins into the air. Tobacco smoke infiltrates indoor spaces, laden with carcinogens and toxins, endangering respiratory health. Poor water quality exacerbates indoor air pollution through the release of volatile compounds and microbial agents during household use. Additionally, the combustion of biomass fuels for cooking emits harmful pollutants, as does the preparation of fast foods, which releases contaminants through high-temperature cooking methods and packaging materials. Fast cards, akin to mosquito killers, also contribute to indoor pollution through the release of chemical compounds and fumes. These products typically contain insecticides or pesticides designed to attract and eliminate mosquitoes and other insects. However, the combustion or vaporization of these chemicals can

introduce harmful pollutants into indoor air, including volatile organic compounds (VOCs) and particulate matter. Prolonged exposure to these pollutants may lead to respiratory irritation, exacerbate existing respiratory conditions, and pose health risks to occupants, particularly those with respiratory sensitivities or compromised immune systems.

1.2 Literature Review:

People are having more intention to maintain their living environment of car to be comfortable. But Outdoor environment is getting worst day by day in the world, due to the pollution emitted by the automobiles and industries. People, who are inhaling the polluted air becomes sicker. Likewise, people who are living in the indoor environment also inhaling worst quality of air. Bad indoor air quality leads to throat, nose and eye irritations, cough, skin itchy, nausea, dizziness, and bad odour. (3)(4)(5) Research on the urban population has confirmed that people spend more than 90% of their daily lifespan in indoor environments. Apart from residential indoor environments, people spend a large proportion of their time in offices, educational institutes, and other different commercial and industrial buildings. Specific research in North America has shown that adults tend to spend 87% of their time in buildings, and the remainder of their time is spent in vehicles (6%) and outdoors (7%) (6). As people spend much of their time in indoor environments, exposure to indoor air pollutants has a significant impact on both human health and effectiveness in the workplace. However, research on air quality has mostly focused on the outdoors, whereas indoor air quality (IAQ) and its impacts have received considerably less attention until the last decade. The energy crisis of the 1970s introduced the importance of energy savings in buildings, which ultimately led to more airtight and insulated buildings worldwide [7]. For energy savings, reduced amounts of fresh air are circulated in air conditioning systems. Moreover, with the advent of improved living standards, more synthetic materials and chemicals are being used in buildings for indoor construction and decorating purposes. Additionally, pesticides, cleaning agents, air fresheners, and gases from cooking are other sources of indoor air pollution.

Because inadequate ventilation, lack of air conditioning systems, human activities, and numerous materials, chemicals, and gases mainly influence indoor pollution, different organizations, such as the United States Environmental Protection Agency (US EPA) and World Health Organization (WHO) have recognized IAQ as a multi-disciplinary phenomenon and classified pollutants into several categories. According to the WHO, in the year 2000, over 1.5 million deaths were caused by indoor air pollution [8]. Moreover, indoor air pollution has been recognized as the third main reason for disability-adjusted life years worldwide [9] Moreover, various studies have been conducted on the health risks of dampness and mold in houses but few studies have been performed in workplaces and schools. The paper of Lanthier-Veilleux et al. [10] is an examination of the independent contribution of residential dampness or mold (i.e., visible mold, mold odor, dampness, or water leaks) to asthma, allergic rhinitis, and respiratory infections among students at the Université de Sherbrooke (Quebec, QC, Canada); while the work of Szulc et al. [11] evaluated the microbiological contamination at a plant biomass processing thermal power station located in Poland.

Among the factors that influence the estimation of human exposure to indoor air pollution,

the pattern of human behavior and activity play a fundamental role. Odeh and Hussein [12] evaluated, for the first time, the human activity pattern of 285 subjects (17–63 years old) residents in Amman(Jordan) in order to use the outcomes in future human exposure studies.

Environmental tobacco smoke (ETS) is also considered a key contributor to indoor air pollution

and public health. In comparison to the large body research on toxicological substances of ETS and concentrations of indoor ETS-dependent PM, less attention has been paid on the correlation between the odor concentration and the chemical composition of ETS. The odor concentrations of field ETS, second-hand smoke (SHS), and third-hand smoke (THS) in prepared samples were determined by Noguchi et al. [13] using the triangle-odor-bag method, while the chemical compositions of the same samples were determined by proton transfer mass spectrometry. Results of this study evidenced that the main

contributing components to the odor of the field ETS samples (acetaldehyde, acetonitrile, acetic acid, and other unknown components with a mass-to-charge ratio (m/z) of 39 and 43) were different from those found in SHS and THS samples.

What's clear, however, is that, just as poorer and marginalized people are disproportionately affected by bad outdoor air quality (14), indoor air pollution is a source of inequality, too. In sub-Saharan Africa, for example, 700,000 people are estimated to have died from indoor air pollution in 2019, many of them from the effects of particles from indoor biomass stoves (15). Cleaner alternatives do exist, but their large-scale use needs a panoply of research-based interventions — from engineering and design to behavioural sciences.

1.3 Rationale of the Study:

The rationale for conducting a study on indoor air pollution (IAP) is multifaceted and underscores the urgent need to address this pervasive health threat. Indoor air pollution poses significant risks to human health, contributing to a range of respiratory diseases and other adverse health outcomes, particularly in developing countries where reliance on solid fuels for cooking and heating is common. Furthermore, IAP disproportionately affects marginalized and low-income populations, exacerbating existing inequalities. Despite its profound impact on public health, indoor air quality has historically received less attention than outdoor pollution, leaving a critical gap in our understanding of this issue. Rapid urbanization, changes in building construction, and modern lifestyles have further exacerbated indoor pollution, necessitating research-based interventions to mitigate its effects.

This study aims to investigate the various sources of indoor air pollution and their impact on human health. The research will examine the link between exposure to indoor air pollutants and respiratory diseases, as well as explore the social and economic factors that contribute to unequal exposure to indoor air pollution.

1.5 Research Questions:

Based on the content provided, here are some potential research questions:

- What are the components playing important role in household physical environment or Household Environment Quality Index (HEQI)?
- What are the disparities in household environment quality in states and population sub groups?
- How does household environment quality, as measured by the Household Environment Quality Index (HEQI), correlate with the prevalence of chronic diseases such as heart disease, lung disease, and cataract?

1.6 Research Objectives.

1. To compute Household Environment Quality Index.
2. To explore the variations of Household Environment Quality Index across states and population groups.
3. To look into the association of selected chronic diseases.

4.5 Chapterization:

I. Introduction

II. Data and methods

III. Understanding Household Environment Quality in India

IV. Linkages of Household Environment Quality Index (HEQI) and Chronic Diseases:

V. Conclusion

VI. References

Chapter 2. Methods & Materials

"Happiness is the highest form of health"

-Dalai Lama

2. Data and Methods

2.1 Data:

We used data from the Longitudinal Ageing Study in India (LASI) conducted from April 2017 to December 2018. The study used stratified, multistage probability cluster random sampling to select 42,949 households. In these households, all individuals aged 45 years and over and their spouses were targeted for interview. The survey was conducted in all states and union territories (hereafter, "states"), but data from Sikkim were not available for this study. The sample was representative of the non-institutionalized population aged 45 years and over (and spouses) at the state level, as well as nationally. The response rate was 95.8% at the household level and 87.3% at the individual level. LASI obtained ethical approval from the Health Ministry's Screening Committee (Government of India) and the Institutional Review Boards at the International Institute for Population Sciences (IIPS) and its collaborating institutions. Written informed consent was obtained from all study participants. We did not prespecify the statistical analysis.

2.2 Indicators:

Table 2.2.1 Descriptive Statistics of Indicators used.

Indicators	Percentage using good	Percentage deprived	Mean	SD	N
cooking fuel	53.85	46.15	0.53	0.49	42949
water facility	96.98	3.02	0.96	0.17	42637
toilet facility	80.16	19.84	0.81	0.39	42737
smoking	75.96	24.04	0.75	0.42	42733
mosquito repellent	4.14	95.86	0.04	0.19	42735
house quality	97.81	2.19	0.97	0.14	42736

Several key indicators were examined to assess household environmental quality among the study participants. The indicators included the type of cooking fuel used, availability of water and toilet facilities,

smoking behaviour, use of mosquito repellent, and overall house quality. Each indicator was evaluated based on the percentage of households categorized as "using good" or "deprived" in relation to the respective environmental factor. The mean and standard deviation (St.dev) for each indicator provided additional insights into the distribution and variability within the dataset. These indicators collectively offer a comprehensive overview of household environmental conditions and serve as foundational elements for further analysis of their associations with health outcomes.

2.3 Methods:

a. Index creation by Principal Component Analysis (PCA):

The indoor pollution index was developed through a systematic approach to ensure robustness and validity. Principal component analysis (PCA) was chosen as the method for selecting indicators due to its ability to reduce the dimensionality of the dataset while retaining important information about the underlying variables. The analysis was performed using STATA, considering six key indicators identified as significant contributors to indoor air pollution: burning of mosquito repellent, dampness in the house, cooking fuel type, sanitation facilities, water quality. The PCA aimed to identify the principal components (PCs) that explained the maximum variance in the dataset, with PCs having eigenvalues greater than or equal to 1 considered for inclusion in the index. Varimax rotation was applied to enhance interpretability and maximize the correlation between the PCs and the indicators. Variables with high factor loadings, indicative of their importance in capturing the variability within each PC, were retained for the indoor pollution index. Furthermore, to address redundancy among highly weighted factors, multivariate correlation coefficients were utilized, ensuring the selection of distinct and non-redundant indicators. This rigorous methodological approach ensured that the indoor pollution index effectively represented the multidimensional nature of indoor air pollution, providing a comprehensive tool for assessing and addressing indoor environmental quality.

b. Validation of Household Indoor Pollution Index Score Using Urbanization and Human Development Index:

To validate the household indoor pollution index score, data on urbanization levels and Human Development Index (HDI) scores were obtained for each state in the study area. Urbanization levels were determined based on the population density and percentage of urban population in the vicinity of each household. HDI scores were calculated using relevant socio-economic indicators, such as education, income, and life expectancy, at the household level. The household indoor pollution indicators was then cross-referenced with the urbanization levels and HDI scores to assess the correlation between these variables. Statistical analyses, including correlation coefficients and regression models, were employed to determine the strength and significance of the relationships. This validation process aimed to evaluate the accuracy and reliability of the household indoor pollution index score in capturing variations in indoor pollution levels associated with urbanization and socio-economic development.

c. Merging Household and Individual file:

The integration of LASI (Longitudinal Aging Study in India) household and individual files was conducted to facilitate comprehensive analysis of indoor pollution and its associated health outcomes at both the household and individual levels. Firstly, the LASI household file, containing information on household characteristics and environmental factors, was merged with the LASI individual file, which includes demographic, health, and socioeconomic data for each household member. The merging process was performed using a unique identifier common to both files, ensuring accurate linkage between households and individuals. Subsequently, variables related to indoor pollution, such as cooking fuel type, sanitation facilities, and housing conditions, were extracted from the integrated dataset for further analysis. Statistical techniques, including descriptive statistics and multivariate regression models, were employed to examine

the relationships between indoor pollution exposures and health outcomes, adjusting for potential confounders.

d. Paired t-test:

A paired t-test analysis was conducted to assess the association between household environment quality index score and health indicators, including lung disease, heart disease, cataract, and diabetes, among individuals in India. The household environment quality index score was calculated based on multiple factors, such as indoor pollution levels, sanitation facilities, and housing conditions, using a standardized scoring system. Health indicators were dichotomously categorized as "Yes" or "No" based on the presence or absence of the respective condition. The null hypothesis assumed that there would be no difference in the means of the household environment quality index score between the "Yes" and "No" groups for each health indicator. Conversely, the alternative hypothesis posited that the mean household environment quality index score would differ between the "Yes" and "No" groups for each health indicator. Statistical significance was assessed using a predetermined alpha level (e.g., $\alpha = 0.05$). The paired t-test provided insights into the strength and direction of the association between household environment quality and health outcomes, thereby informing potential interventions to improve indoor environmental conditions and mitigate health risks.

e. Models:

We utilized generalized linear models (GLMs) with a logistic regression function as the analytical tool. Logistic regression is a specific type of GLM well-suited for analyzing relationships between categorical dependent variables, like the presence or absence of heart disease and cataracts, and independent variables that can be categorical or continuous. In this case, our dependent variable is binary (yes/no), and some

independent variables are categorical. Logistic regression estimates the odds of the dependent variable occurring for a given set of independent variables. The logistic regression model expresses the relationship between the independent variables (X) and the probability (P) of the dependent variable (Y) occurring using a sigmoid function. The sigmoid function takes a linear combination of the weighted independent variables ($\beta_0 + \beta_1X_1 + \beta_2X_2 + ... + \beta_nX_n$) and transforms it into a probability between 0 and 1.

Here's the mathematical equation for the logistic regression model:

$$P(Y = 1 | X) = 1 / (1 + e^{(-(\beta_0 + \beta_1X_1 + \beta_2X_2 + ... + \beta_nX_n))})$$

- $P(Y = 1 | X)$ represents the probability of the dependent variable (Y) being 1 (event occurring) given the set of independent variables (X).
- β_0 is the intercept term.
- β_1 to β_n are the regression coefficients for each independent variable (X_1 to X_n).
- e is the base of the natural logarithm (approximately 2.718).

The regression coefficients (β) indicate the direction and strength of the association between each independent variable and the log-odds of the dependent variable. Positive coefficients suggest an increased likelihood of the event (Y = 1) with increasing values of the independent variable, while negative coefficients suggest a decreased likelihood.

Chapter 3.

Insights into Household Environment Quality and Socio-Economic Indicators: A Comprehensive Analysis

"As we encounter each other, we see our diversity — of background, race, ethnicity, belief – and how we handle that diversity will have much to say about whether we will in the end be able to rise successfully to the great challenges we face today."

— Dan Smith

Household Environment Quality Index:

Table 3.1 Eigenvalues and Proportion of Variance Explained by Principal Components

Components	Eigen value	Difference	Proportion	Cumulative
Comp1	2.76036	0.652743	0.23	0.23
Comp2	2.10761	0.195087	0.1756	0.4057
Comp3	1.91253	0.0214544	0.1594	0.565
Comp4	1.89107	0.0648906	0.1576	0.7226
Comp5	1.82618	0.595892	0.1522	0.8748
Comp6	1.23029	0.984299	0.1025	0.9773
Comp7	0.245991	0.220965	0.0205	0.9978
Comp8	0.0250253	0.024465	0.0021	0.9999
Comp9	0.000560283	0.000354579	0	1
Comp10	0.000205705	3.20403E-05	0	1
Comp11	0.000173664	0.000173664	0	1
Comp12	0	.	0	1

In the Principal Component Analysis (PCA) results presented, a total of 12 components were extracted from the dataset. The amount of variation explained by each component is indicated by its eigenvalue, which also serves as a measure of the difference in eigenvalues across components, the percentage of variance explained, and the cumulative percentage of variance explained up to that component.

With an eigenvalue of 2.76036, the first principal component (Comp1) has the greatest value and can account for a significant amount of the variation in the data. This component has a cumulative percentage of 23% and explains 23% of the variation. Decreasing shares of variation are explained by subsequent components, even if they still contribute to the overall variance. Comp2, for example, accounts for 17.56% of the variance adding up to a total percentage of 40.57%.

The eigenvalues decline with increasing component complexity, indicating fractions of explained variance that are getting smaller. All the same, every part adds to our comprehension of the data structure. Comp10,

Comp11, and Comp12 are the final components, and their contribution to the cumulative proportion of variance becomes insignificant. Their eigenvalues are nearly zero, suggesting limited variance explained.

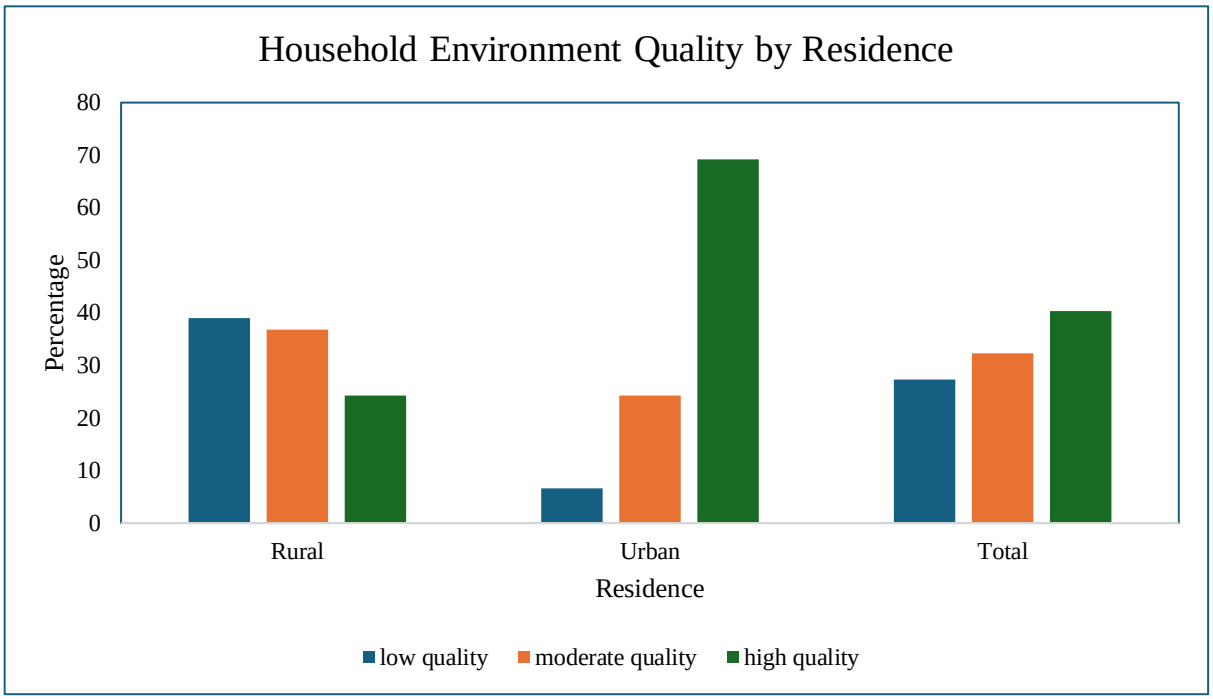
Table 3.2 Socio-economic and Environmental Indicators and Their Scores

Indicators	Score
water quality deprived	-0.1188
water quality good	0.1328
No dampness	0.0274
Dampness present	0.0021
fast card not using	0.0598
fast card using	-0.034
toilet facility good	0.4411
toilet facility deprived	-0.4333
smoking using	-0.2027
smoking not	0.2123
cooking fuel good	0.4971
cooking fuel bad	-0.4971

The analysis reveals a clear hierarchy of factors influencing household environment quality. At the top of the pyramid sit sanitation facilities, particularly good toilets (0.4411) and the use of clean cooking fuel (0.4971). These factors boast the strongest positive scores, underlining their critical role in creating a healthy and safe home environment. Access to clean water, though scoring moderately positive (0.1328), also emerges as a crucial element. Conversely, deprived water quality (-0.1188) and the absence of proper toilet facilities (-0.4333) have the most significant negative influence, highlighting their detrimental impact on the overall environment.

Beyond sanitation and water access, the analysis emphasizes the importance of healthy living habits. Smoking habits come into focus, with not smoking (0.2123) contributing positively while smoking (-0.2027) exerts a negative influence. This underscores the well-documented health risks associated with smoking within the home environment. The quality of the house itself seems to play a less prominent role compared to sanitation, water, and healthy habits. While good house quality receives a positive score (0.0274), its impact is less pronounced compared to the top factors. Similarly, deprived house quality has a minimal score (0.0021). This suggests that the overall structure of the house may be less critical for immediate well-being compared to access to clean water, proper sanitation, and healthy living choices.

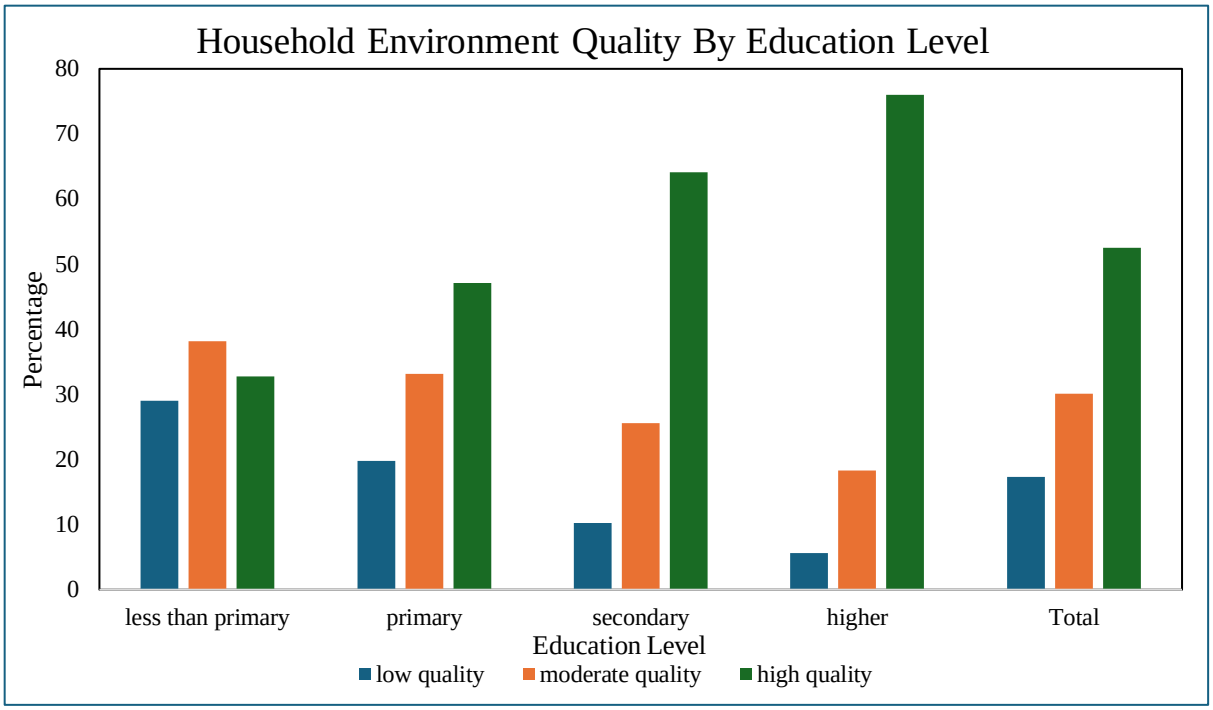
Fig 3.1 Variations in Household Environment Quality by residence



The graph depicts a disparity in Household Environment Quality Index (HEQI) between rural and urban areas. A higher proportion of rural households reside in low HEQI conditions, whereas a greater percentage of urban households enjoy high HEQI. This suggests a positive correlation between urbanization and HEQI. Additionally, urban areas likely benefit from superior access to resources like sanitation and waste

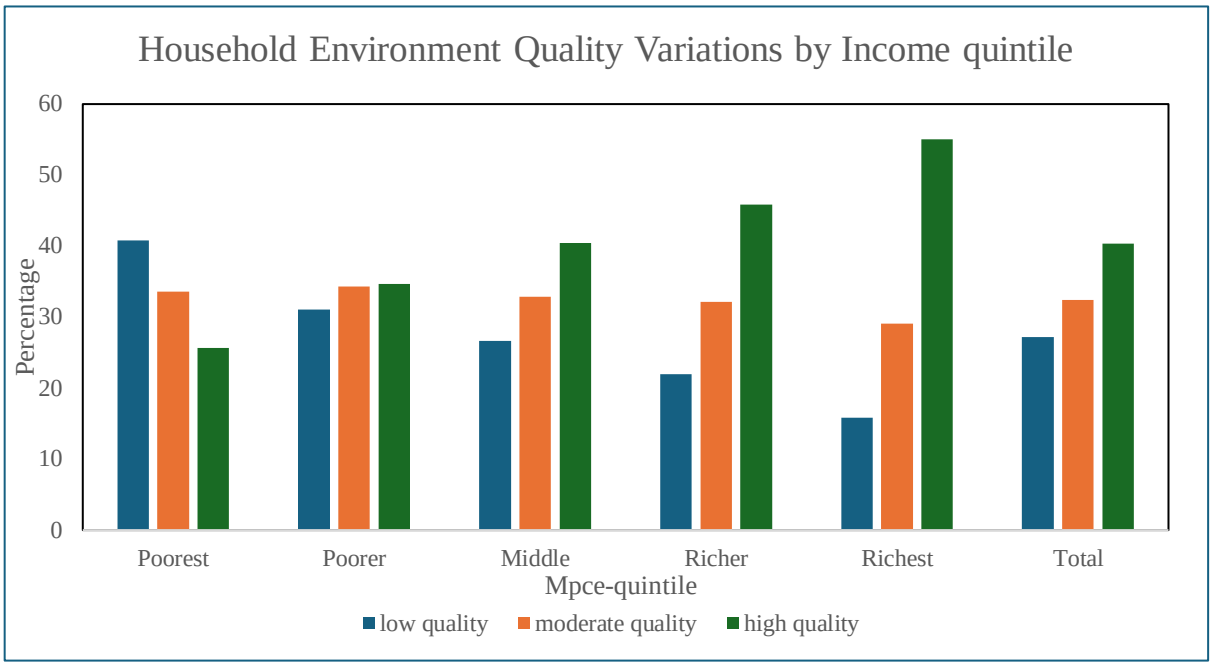
disposal, further improving HEQI. Finally, higher incomes in urban areas might allow residents to afford better housing and amenities, contributing to a higher overall HEQI.

Fig 3.2 Variations in Household Environment Quality by Education Level



The graph shows that there is a positive correlation between educational level and household environment quality. As educational level increases, the percentage of people with a high household environment quality also increases. Conversely, the percentage of people with a low household environment quality decreases as educational level increase. People with higher levels of education may have higher incomes, which allows them to afford better housing in better neighbourhoods. Additionally, people with higher levels of education may be more aware of the importance of a healthy home environment and may be more likely to take steps to improve the quality of their home environment.

Fig 3.4 Household Environment Quality Variations by Income Quintile



The graph suggests a positive correlation between income and household environment quality. As income increases, the percentage of people with a high household environment quality also increases. Conversely, the percentage of people with a low household environment quality decrease as income rises. This trend likely reflects several factors. Higher income allows people to afford better housing in neighbourhoods with better environmental quality. Additionally, people with more income may have greater resources to improve their home environment.

Chapter 4.

State-Level Analysis of Socioeconomic Determinants and Household Environment Quality in India

" It is the health that is the real wealth and not pieces of Gold and Silver."

--Mahatma Gandhi

Table 4.1 Correlational Analysis of Deprivation, Urbanization and Human Development Factors at the state level

Prevalence of Indicators	HDI	Urbanization
fast card using	-0.0444	-0.0676
toilet facility deprived	-0.6536	-0.3571
smoking using	-0.0164	-0.135
cooking fuel bad	-0.6819	-0.7925
water quality deprived	0.0205	-0.194
dampness in house	0.1696	0.0535

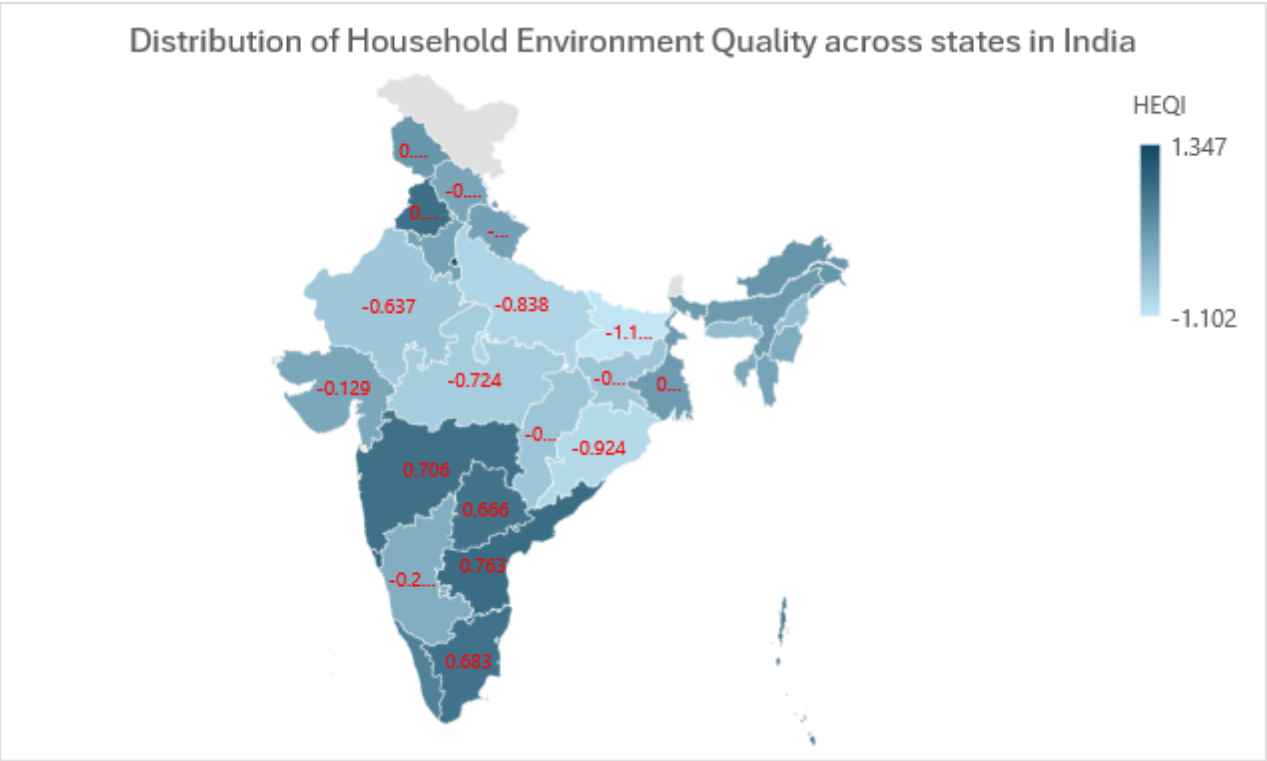
A comprehensive analysis of various indicators such as water quality, house quality, access to fast transportation, toilet facilities, smoking prevalence, and cooking fuel quality, significant correlations emerge. Notably, factors such as toilet facility deprivation and poor cooking fuel quality exhibit strong negative correlations with both the Human Development Index (HDI) and urbanization rates.

Table 4.2 Correlational Analysis of Household Environment Quality Index (HEQI) and Chronic Diseases

Indicator	HEQI
HDI	0.79
Urbanization	0.78
Prevalence of Cataract	-0.15
Prevalence of Diabetes	-0.03
Prevalence of Hypertension	-0.14
Prevalence of lung disease	-0.04
Prevalence of heart disease	0.22

The provided indicators represent the correlation coefficients with the Household Environment Quality Index (HEQI), offering insights into their respective associations. negative correlations are observed between HEQI and other health conditions, including cataract (-0.15), diabetes (-0.03), hypertension (-0.14), and lung disease (-0.04), suggesting that as HEQI improves, the prevalence of these conditions tends to decrease.

Fig 4.1 Distribution of Household Environment Quality across states in India.



The map reveals a potential disparity in household environment quality (HEQI) across different states in India. States like Uttar Pradesh, Rajasthan, and Madhya Pradesh appear to have a larger portion of their population residing in areas with lower HEQI, indicating potentially lower quality housing environments. Conversely, states concentrated in the southern and eastern parts of India seem to have a higher share of their population in areas with higher HEQI, suggesting generally better household environments.

Table 4.3 Socioeconomic and Environmental Determinants of Human Development: A State-Level Analysis in India

state	HDI	Urbanization	smoking inside	unimproved toilet	bad cook fuel	bad water	fast card	dampness	HEQI
Andaman and Nicobar	0.707	37.700	0.102	0.065	0.371	0.072	0.041	0.003	0.561
Andhra Pradesh	0.648	33.360	0.222	0.152	0.156	0.008	0.013	0.009	0.763
Arunachal Pradesh	0.684	22.940	0.131	0.058	0.514	0.167	0.077	0.023	0.142
Assam	0.651	14.100	0.149	0.038	0.674	0.040	0.153	0.042	0.061
Bihar	0.551	11.290	0.149	0.557	0.684	0.005	0.014	0.003	-1.102
Chandigarh	0.827	97.250	0.240	0.009	0.020	0.008	0.038	0.029	1.347
Chhatisgarh	0.629	23.240	0.195	0.237	0.749	0.018	0.023	0.058	-0.598
Delhi	0.839	97.500	0.229	0.041	0.017	0.011	0.069	0.015	1.299
Goa	0.806	62.170	0.075	0.132	0.198	0.019	0.034	0.016	0.855
Gujarat	0.698	42.600	0.274	0.236	0.498	0.020	0.055	0.012	-0.129
Haryana	0.724	34.880	0.525	0.090	0.503	0.015	0.079	0.029	-0.055
Himachal Pradesh	0.761	10.030	0.354	0.094	0.528	0.123	0.022	0.024	-0.107
Jammu and Kashmir	0.663	27.380	0.440	0.120	0.382	0.045	0.035	0.005	0.124
Jharkhand	0.618	24.050	0.103	0.317	0.718	0.011	0.045	0.013	-0.598
Karnataka	0.706	38.670	0.202	0.335	0.470	0.008	0.044	0.030	-0.232
Kerala	0.775	47.700	0.170	0.022	0.462	0.048	0.029	0.045	0.457
Madhya Pradesh	0.616	27.630	0.318	0.352	0.641	0.005	0.045	0.035	-0.724
Maharashtra	0.750	45.220	0.169	0.136	0.238	0.014	0.063	0.009	0.706
Manipur	0.686	29.210	0.327	0.004	0.505	0.358	0.000	0.022	-0.195
Meghalaya	0.704	20.070	0.406	0.027	0.712	0.160	0.031	0.000	-0.453
Mizoram	0.747	52.110	0.690	0.012	0.413	0.159	0.056	0.000	-0.069
Nagaland	0.678	28.860	0.164	0.029	0.755	0.272	0.024	0.003	-0.459
Odisha	0.649	16.690	0.138	0.437	0.726	0.029	0.058	0.001	-0.924
Puducherry	0.752	68.330	0.052	0.131	0.148	0.001	0.019	0.026	0.996
Punjab	0.738	37.480	0.161	0.041	0.319	0.007	0.035	0.176	0.731
Rajasthan	0.638	24.870	0.438	0.330	0.557	0.012	0.018	0.006	-0.637
Tamil Nadu	0.738	48.400	0.099	0.222	0.177	0.012	0.009	0.020	0.683
Telangana	0.705	33.000	0.214	0.217	0.145	0.013	0.038	0.016	0.666
Tripura	0.667	26.170	0.520	0.025	0.641	0.015	0.200	0.000	-0.127
Uttar Pradesh	0.592	22.270	0.304	0.498	0.545	0.003	0.024	0.003	-0.838
Uttarakhand	0.758	30.230	0.425	0.079	0.526	0.012	0.008	0.027	-0.025
West Bengal	0.674	31.870	0.356	0.111	0.504	0.007	0.093	0.025	0.067

This comprehensive state-level analysis in India explores the intricate interplay between various socioeconomic and human development determinants and their impact on household environment quality.

The study investigates key indicators such as the Human Development Index (HDI), urbanization rates, prevalence of smoking indoors, access to improved sanitation facilities, quality of cooking fuel, water quality, fast card usage, and household dampness, alongside the Household Environment Quality Index (HEQI). Findings reveal significant disparities among states, with higher HDI and urbanization generally associated with improved household environment quality. However, disparities persist, with some states exhibiting concerning trends such as higher rates of indoor smoking and inadequate access to essential amenities like improved toilets and clean water.

Chapter 5: ***Linkages of Household Environment Quality Index (HEQI) on Chronic Diseases: Insights from Regression Analyses***

"In a gentle way you can shake the World"

-----Mahatma Gandhi

The investigation into the association between the Household Environment Quality Index (HEQI) and chronic diseases unveils a critical link between living conditions and health outcomes. Through rigorous analysis, it becomes evident that a higher HEQI is significantly correlated with reduced prevalence and risk of chronic diseases. Improved household environments, characterized by better air quality, access to clean water, sanitation facilities, and overall hygiene, appear to contribute to a lower incidence of chronic illnesses such as cardiovascular diseases, respiratory disorders, and certain cancers. These findings underscore the importance of addressing environmental factors within households as a crucial component of preventative healthcare strategies.

Table 5.1 Paired t test of Chronic Diseases with HEQI

	HEQI			
Chronic Diseases	Mean Diff.	p-value	t-value	N
Heart Disease	0.572	0	16.706	70967
Lung Disease	0.288	0	3.731	70965
Cataract	0.256	0	13.21	33057
Hypertension	0.554	0	40.34	70960
Arthritis	-0.05	0.067	-1.574	10013
Diabetes	0.869	0	46.21	70957

Null Hypothesis (H0): There is no significant difference in mean Household Environment Quality Index (HEQI) scores between individuals with and without the respective chronic diseases.

Alternative Hypothesis (H1): There is a significant difference in mean HEQI scores between individuals with and without the respective chronic diseases.

The table presents the results of paired t-tests examining the association between HEQI and various chronic diseases. Each row corresponds to a different chronic disease, with accompanying statistics including the mean difference in HEQI scores (Mean Diff.), the p-value, the t-value, and the sample size (N). Significant mean differences in HEQI scores are observed for heart disease, lung disease, cataract, hypertension, and

diabetes, as evidenced by p-values of 0.000 (indicating statistical significance at the conventional alpha level of 0.05). These findings suggest that individuals affected by these chronic diseases tend to have significantly different HEQI scores compared to those without the conditions. Conversely, no significant difference in HEQI scores is observed for arthritis, as indicated by a p-value of 0.067, suggesting that household environment quality may not significantly differ between individuals with and without arthritis.

5.2 How Household Environment Quality Index is impacting heart disease.

	Heart Disease				
Independent Variables	OR	z-value	p-value	95% CI	
food	0.006	0.140	0.892	-0.079	0.090
mpce_quintile					
2 Poorer	-0.020	-0.260	0.798	-0.175	0.135
3 Middle	0.153**	2.020	0.043	0.005	0.301
4 Richer	0.235	3.190	0.001	0.091	0.380
5 Richest	0.483	6.740	0.000	0.342	0.623
residence					
2 Urban	0.348	7.610	0.000	0.258	0.437
ht002					
2 No	-1.427	-32.370	0.000	-1.513	-1.340
HEQI_c					
moderate	0.260	1.640	0.100	-0.050	0.570
good	0.405	2.660	0.008	0.107	0.703
caste					
Scheduled Caste (SC)	0.648	6.750	0.000	0.460	0.836
backward class (OBC)	0.697	8.100	0.000	0.528	0.866
None of above	0.914	10.550	0.000	0.744	1.084
_cons	-3.956	-21.330	0.000	-4.320	-3.593

Note: '' at 10 % significance level, '**' at 5% significance level, '***' at 1% significance level*

In the logistic regression analysis focusing on heart disease, the Household Environment Quality Index (HEQI) emerges as a significant factor. With "poor" HEQI as the reference category, individuals with "moderate" HEQI levels display a trend towards lower odds of heart disease (OR = 0.260, p = 0.100), although this association does not reach statistical significance at conventional levels. Conversely, individuals with "good" HEQI levels exhibit a significant reduction in the odds of heart disease compared to those with "poor" HEQI (OR = 0.405, p = 0.008). These findings suggest that household environments categorized as "good" HEQI are associated with lower odds of heart disease occurrence relative to those categorized as "poor" HEQI.

5.2 How Household Environment Quality Index is impacting Cataract.

	Cataract				
Independent Variables	OR	z-value	p-value	95% CI	
diabetes					
2 No	-0.28638	-8.29	0	-0.35405	-0.2187
food	-0.25309	-9.37	0	-0.30603	-0.20016
mpce_quintile					
2 Poorer	-0.2617	-6.09	0	-0.34595	-0.17745
3 Middle	-0.40338	-9.44	0	-0.48713	-0.31963
4 Richer	-0.47303	-11.16	0	-0.55608	-0.38997
5 Richest	-0.64543	-14.92	0	-0.73024	-0.56062
residence					
2 Urban	-0.15512	-5.39	0	-0.21152	-0.09871
HEQI_c					
moderate	-0.22324	-2.88	0.004	-0.37537	-0.07112
good	-0.47824	-6.43	0	-0.62407	-0.33241
caste					
Scheduled Caste (SC)	0.432261	8.19	0	0.328789	0.535734
backward class (OBC)	0.54677	11.59	0	0.454326	0.639214
None of above	0.434695	8.9	0	0.338971	0.530419
_cons	0.039719	0.4	0.692	-0.15698	0.236417

In the logistic regression analysis focusing on cataract occurrence, several significant determinants emerge, shedding light on the multifaceted nature of factors influencing cataract risk within the population. Notably, individuals residing in households with "good" levels of Household Environment Quality Index (HEQI) demonstrate a substantial decrease in the odds of cataract compared to those in households with "moderate" HEQI levels (OR = 0.478, $p < 0.001$), suggesting a protective effect of improved household environmental quality against cataract development. Moreover, socioeconomic status, as indicated by mpce_quintile, exhibits a consistent pattern, with individuals in higher income categories showing progressively reduced odds of cataract relative to the "poorest" category (ORs ranging from -0.262 to -0.645, all $p < 0.001$). Additionally, urban residence is associated with decreased odds of cataract compared to rural areas (OR = -0.155, $p < 0.001$). Conversely, certain caste categories, such as Scheduled Caste and Other Backward Class, are associated with elevated odds of cataract compared to the reference category.

Chapter 6: ***Conclusion & References***

6. Conclusion

The background established the significance of indoor air pollution as a critical health concern, particularly in developing countries where reliance on solid fuels for cooking contributes to respiratory diseases. The literature review highlighted the increasing recognition of indoor air quality as a research focus and the diverse sources of indoor pollution affecting human health. Insights into household environment quality and socio-economic indicators revealed the complex interplay between various factors and their impact on indoor pollution. State-level analysis in India unveiled disparities in household environment quality across regions, with socio-economic factors and urbanization playing significant roles. Finally, the association between household environment quality and chronic diseases underscored the importance of addressing indoor pollution for public health outcomes. These findings collectively emphasize the urgent need for targeted interventions to improve household environment quality and mitigate the burden of chronic diseases, particularly in marginalized communities.

7. References

1. Lim SS, Vos T, Flaxman AD, Danaei G, Shibuya K, Adair-Rohani H, et al. A comparative risk assessment of burden of disease and injury attributable to 67 risk factors and risk factor clusters in 21 regions, 1990-2010: A systematic analysis for the Global Burden of Disease Study 2010. *Lancet.* 2012;380:2224–60. [PMC free article] [PubMed] [Google Scholar]
2. Cincinelli A, Martellini T. Indoor air quality and health. Int J Environ Res Pu. 2017;14:1286.
3. Review of Research On Air conditioning System And Indoor Air Quality Control For Human Health
4. Towards an integrative approach of improving indoor air quality
5. Experimental Study on Thermal Properties of Nano-TiO_2 Embedded Paraffin (NEP) for Thermal Energy Storage Applications
6. Vilčeková, S.; Apostoloski, I.Z.; Mečiarová, Ľ.; Burdová, E.K.; Kiseľák, J. Investigation of indoor air quality in houses of Macedonia. *Int. J. Environ. Res. Public Health* **2017**, *14*, 37. [**Google Scholar**] [**CrossRef**] [**Green Version**]
7. Prajakta, P. Shrimandilkar, Indoor Air Quality Monitoring for Human Health. *Ijmer* **2013**, *3*, 891–897. Available online: **http://www.ijmer.com/papers/Vol3_Issue2/BV32891897.pdf** (accessed on 18 January 2020).
8. World Health Organization. *Indoor Air Pollution: National Burden of Disease Estimates*; WHO: Geneva, Switzerland, 2007; Available online: **https://www.who.int/airpollution/publications/indoor_air_national_burden_estimate_revised.pdf?ua=1** (accessed on 5 June 2019).
9. Apte, K.; Salvi, S. Household air pollution and its effects on health. *F1000Research* **2016**, *5*, 2593. [**Google Scholar**] [**CrossRef**]
10. Lanthier-Veilleux, M.; Baron, G.; Généreux, M. Respiratory Diseases in University Students Associated with Exposure to Residential Dampness or Mold. Int. J. Environ. Res. Public Health 2016,13, 1154. [CrossRef] [PubMed]
11. 6. Szulc, J.; Otlewska, A.; Okrasa, M.; Majchrzycka, K. Microbiological Contamination at Workplaces in a Combined Heat and Power (CHP) Station Processing Plant Biomass. Int. J. Environ. Res. Public Health 2017,14, 99. [CrossRef] [PubMed]
12. 7. Odeh, I.; Hussein, T. Activity Pattern of Urban Adult Students in an Eastern Mediterranean Society. Int. J. Environ. Res. Public Health 2016, 13, 960. [CrossRef] [PubMed]
13. 8. Noguchi, M.; Tanaka, S.; Watanabe, K.; Yamasaki, A. Correlation between Odor Concentration and Volatile Organic Compounds (VOC) Composition of Environmental Tobacco Smoke (ETS). Int. J. Environ. Res. Public Health 2016, 13, 994. [CrossRef] [PubMed]
14. A. Jbaily *et al. Nature* **601**, 228–233; 2022
15. Khavari et al. Nature Sustain. https://doi.org/grqrbg; 2023